

Improving Object Detection in Hazy Setting: Using AOD-Net, DCP and variants of R-CNN

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Abstract

Single image haze removal is an ill-posed problem. Several methods have been proposed that use various constraints/priors to get plausible dehazing solutions. The key to achieve haze removal is to estimate a medium transmission map for an input hazy image. In this paper, we analyze several dehazing techniques and state-of-the-art object detection algorithms in order to improve object detection performance in hazy setting. AOD-net is one of the primary approaches that we have considered to enhance the image quality for object detection. Apart from AOD-Net, we have used other approaches such as DCP and Histogram equalization whose yield was not visually appealing but performed much better when used with the object detection algorithms.

Similar to dehazing, numerous algorithms have been proposed for object detection with the state-of-the-art algorithms such as Faster R-CNN, YOLO, RetinaNet and Mask-RCNN which perform extremely well. But, all these algorithms have been trained and tested on quality images and might not perform well on degraded images. Our goal is to create an end-to-end system which will take care of image processing to enhance image semantics and use the resultant image for an improved object detection. In this paper, we have explored few different algorithms for object detection and compared their performance. Details of all such approaches have been discussed in the following sections.

1. Introduction

Haze is an atmospheric phenomenon where dust, smoke, and other dry particles obscure the clarity of the atmosphere. This results in a loss of contrast in the subject of the image, due to the effect of light scattering through the haze particles. These degradation in turn may affect the performance of many computer vision systems such as classifica-

tion and detection. To overcome the degradation caused by haze, image and video-based haze removal algorithms have been proposed in the literature[7][17][1][15][6][16][13]. The removal of haze from the images, videos is of great importance in this era. Much of this can be attributed to its profound applications in traffic monitoring, autonomous driving, video surveillance, satellite imaging and many more.

Haze removal is a challenging problem because the haze transmission depends on the unknown depth which varies at different positions. The key idea of the dehazing process is to estimate the atmospheric light magnitude along with the recovery of the transmission matrix. Various methods have been employed to remove the haze from a single image. They can be summarized as follows. First one is Histogram based[6] where the histogram of grey levels(GL) in a window around each pixel is generated first. The cumulative sum of GL's over the histogram is used to map the input pixel GLs to output GLs. If a pixel has a GL lower than all others in the surrounding window the output is maximally black; if it has the median value then the output is 50% gray. Contrast based[17] method employs adaptive image-contrast enhancement based on a generalization of histogram equalization for the dehazing task, and third one is saturation[2] based method in which if the saturation is less than target saturation, a filtering function is selected, operable on the image to enhance saturation as a function of the average saturation and desired saturation thereby enhancing the image. In addition, methods using multiple images have also been proposed. For example, polarization based method[16] which attempts to remove the haze effect through multiple images taken with different degrees of polarization.

The hazy image formation can be described in terms of atmospheric scattering model, which can be formally written as below.

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (1)$$

where $I(x)$ is the observed hazy image, $J(x)$ is the real

scene to be recovered, $t(x)$ is the medium transmission, A is the global atmospheric light, and x indexes pixels in the observed hazy image I . Transmission map is the distance-dependent factor that affects the fraction of light that reaches the camera sensor. When the atmospheric light A is homogeneous, the transmission map can be expressed as $t(x) = e^{-\beta d(x)}$, where β represents attenuation coefficient of the atmosphere and $d(x)$ is the scene depth. In single image dehazing, given I , the goal is to estimate J .

It can be observed from Eq. 1 that there exists two important aspects in the dehazing process: (1) accurate estimation of transmission map, and (2) accurate estimation of atmospheric light. Apart from several works that focus on estimating the atmospheric light[12], most of the other algorithms concentrate more on the accurate estimation of the transmission map and they leverage empirical rule in estimating the atmospheric light [5]. This is mainly due to the common belief that good estimation of transmission map will lead to better dehazing. These methods can be broadly divided into two main groups: prior based methods and learning-based methods. Prior-based methods often leverage different priors in characterizing the transmission map such as dark-channel prior[5], while learning based methods are based on convolution neural networks that are recently proposed for low-level vision tasks of image restoration/reconstruction and they have been modified for the dehazing task as well. The learning based methods attempt to learn the transmission map directly from the training data[1][7][18]

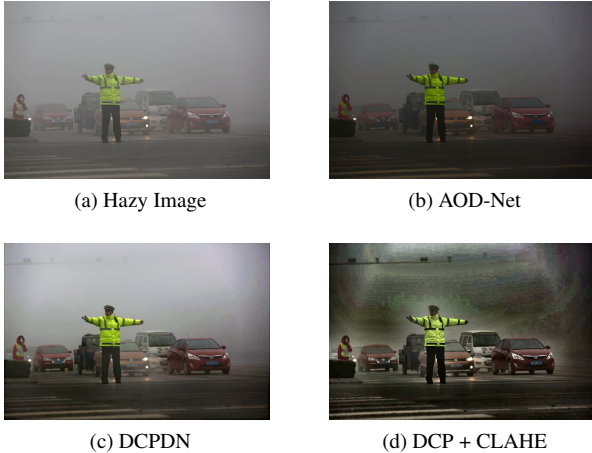


Figure 1: Sample Hazy image and dehazed results using the AOD-Net, DCPDN, and proposed DCP+CLAHE method

Dehaze-net is a state of the art of the dehazing algorithm based on a novel deep learning approach. Dehazenet explicitly learns the mapping relations between raw hazy images and their associated medium transmission maps and hence

performs very well in the dehazing task. AOD-Net[7] uses the re-formulation of the $t(x)$ to integrate $t(x)$ and A into one new variable. As a result, their convolutional dehazing model was fully end-to-end i.e. $J(x)$ was directly generated from $I(x)$, without any intermediate parameter estimation step. The latest DCPDN network(Analysis is done in the later sections) architecture[18] consists of four modules: 1) Pyramid densely connected transmission map estimation net, 2) Atmosphere light estimation net, 3) Dehazing via reformulated Eq. 1, and 4) Joint discriminator for discriminating between high/low confidence transmission map.

The remainder of this paper is organized as follows. In Section II, we review some of the joint models and Object detection algorithms, which provides background knowledge to understand the design of Object detection algorithms that have been implemented/analysed in this paper. In Section III, we present details of the proposed joint models which enhance the detection performance. Experiments are presented in Section IV followed by conclusions in Section V.

2. Related Works

Several algorithms have been devised for the object detection in an image. The R-CNN[3] method trains CNNs to classify the proposal regions into object categories or background. R-CNN mainly plays as a classifier, and it does not predict object bounds. Its accuracy depends on the performance of the region proposal module. Faster R-CNN[14] is a state-of-the-art object detection network in which authors introduce a Region Proposal Network that shares full-image convolutional features with the detection network. Thus Faster R-CNN enables a unified, deep-learning-based object detection system and the learned RPN also improves region proposal quality and thus the overall object detection accuracy. The above approaches are applied to a sparse set of candidate object locations. In contrast, one-stage detectors that are applied over a regular, dense sampling of possible object locations have the potential to be faster and simpler, but have trailed the accuracy of two-stage detectors thus far. RetinaNet[9], addresses the foreground-background class imbalance encountered during training of dense detectors by introducing a novel loss function called Focal Loss. Focal loss focuses training on a sparse set of hard examples and prevents the vast number of easy negatives. In this project we jointly tune AOD-Net with RetinaNet and evaluate the performance. The latest advancement in the field of detection is a new algorithm called Mask RCNN[4]. Mask R-CNN, extends Faster R-CNN by adding a branch for predicting an object mask in parallel with the existing branch for bounding box recognition. We have presented an extensive analysis of performance of Mask RCNN for the dehazing tasks in the later sections.

2.1. Object detection in a Hazy environment

As we can observe in the above sections, very little attention has been dedicated towards object detection under hazy conditions. This is a very important problem because, in autonomous driving, an automated car requires a robust detection of road lanes, traffic lights, and other traffic agents not only in a modest weather condition but also in the presence of fog/smoke. In this project, we specifically try to build algorithms which improve semantic understanding of the image/video measured using Mean Average Precision(mAP) criterion.

AOD-Net[7] is a state-of-the-art approach, where the three stage process of dehazing the image was reduced to 2-stage. This modification reduced the computation time and the pixel-domain reconstruction error. In the AOD-Net[7] paper, the authors have coupled AOD-Net with multiple object detection algorithms such as [14][11] for the object detection. The AOD-Net[7] paper also does a comparative study of coupling different dehazing algorithms[7][5][1] with Faster-RCNN[14].

In this project we will explore newer state-of-the-art object detection algorithms: RetinaNet[9] and Mask-RCNN[4]. These algorithms perform much better compared to Faster R-CNN and we would like to see whether we can achieve better performance compared to JAOD-Faster RCNN model.

3. Proposed Methods

In this project we compare several combinations of dehazing and object detection algorithms. Firstly, we start off with building AOD-Net[7]. The architecture of the AOD-Net is as follows. Lets first rewrite the formula (1) as below.

$$\frac{1}{J(x)} = \frac{1}{t(x)}I(x) - A\frac{1}{t(x)} + A \quad (2)$$

AOD-Net's core idea is to unify the two parameters $t(x)$ and A into one formula, i.e. $K(x)$ in (3), and to directly minimize the pixel-domain reconstruction errors. The unified formula can be written as

$$J(x) = K(x)I(x)K(x) + b; \text{ where}$$

$$K(x) = \frac{\frac{1}{t(x)}(I(x) - A) + (A - b)}{I(x) - 1} \quad (3)$$

The AOD-Net consists of two modules, as illustrated in Figure 2: a K-estimation module that estimates (3), followed by a clean image generation module that utilizes $K(x)$ as its input-adaptive parameters to estimate $J(x)$

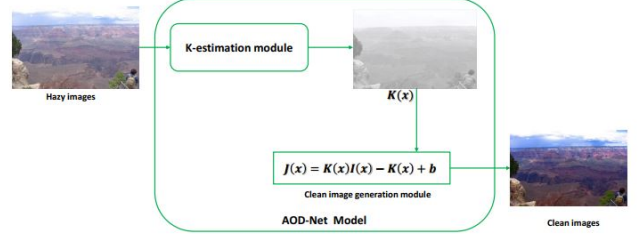


Figure 2: Diagram of AOD-Net

Similar to the original AOD-Net[7] implementation, in our model we use five convolutional layers and form multi-scale features by fusing varied size filters. The "concat1" layer of AOD-Net concatenates features from the layers "conv1" and "conv2". Similarly, "concat2" concatenates those from "conv2" and "conv3"; "concat3" concatenates those from "conv1", "conv2", "conv3", and "conv4". Such a multiscale design captures features at different scales, and the intermediate connections also compensate for the information loss during convolutions. Following the K-estimation module, the clean image generation module consists of an element-wise multiplication layer and several element-wise addition layers, in order to generate the recovered image via calculating (3). The layers are added as shown in Figure 3.

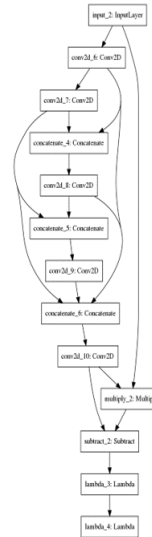


Figure 3: Model Architecture of AOD-Net

Our contribution in this paper is a joint AOD-Net + RetinaNet network architecture which is illustrated in Figure 4 below. RetinaNet is a one-stage object detector based on Feature Pyramid Network with the cross-entropy loss replaced by Focal loss[9].

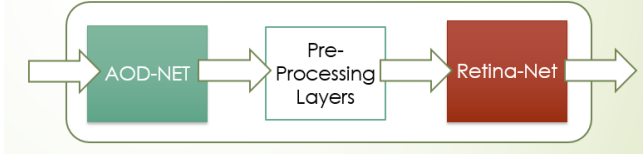


Figure 4: Model Architecture of AOD-Net + RetinaNet

The network essentially consists of the following three modules: 1) AOD-Net which performs the dehazing task, 2) Preprocessing layer which processes the dehazed images to make it compatible with RetinaNet architecture as its built on top of ResNet backbone. In the preprocessing layer, the image is converted to BGR format and each channels are subtracted with their mean values(see Figure 5). The preprocessing layer also helps in enhancing classifier’s accuracy. In naive RetinaNet model preprocessing was taken care by built-in classes, but in the joint model, custom Lambda layers have been added to take care of the preprocessing steps, 3) RetinaNet processes the dehazed images and produces bounding boxes across the objects in the images. The results of this experiment is described in detail in the upcoming section.

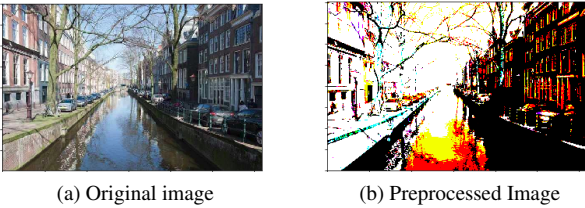


Figure 5: Original image and Preprocessed image

Our second contribution in this paper is a combination of CLAHE, Dark Channel Prior[5] and RetinaNet for enhanced detection. Ordinary Adaptive Histogram Equalization (AHE) tends to over-amplify the contrast in near-constant regions of the image, since the histogram in such regions is highly concentrated. As a result, adaptive histogram equalization may cause noise to be amplified in near-constant regions. Contrast Limited AHE (CLAHE) is a variant of adaptive histogram equalization in which the contrast amplification is limited, so as to reduce this problem of noise amplification. The dark channel prior is based on the statistics of the outdoor images. It is based on a key observation - most local patches in haze-free outdoor images contain some pixels which have very low intensities in at least one color channel. Using this as a prior with the haze imaging model, we can directly estimate the thickness of the haze and recover a high quality haze-free image. Applying the prior into the haze imaging model, single im-

age haze removal becomes simpler and more effective. We apply CLAHE and DCP to a hazy image to recover the dehazed image. This dehazed image is then passed through RetinaNet for the object detection. Our experiments show that this method outperforms all state-of-the-art methods described currently in the literature.

Our third contribution to this paper is a combination of two state-of-the-art novel architectures namely DCPDN[18] and Mask R-CNN[4] for object detection in hazy setting. We first obtain dehazed images using pre-built DCPDN model. The images are then passed to Mask R-CNN for the object detection. We observe that for a subset of the RTTS dataset[8], the detection capability was much better as shown in Figure 6. Although overall mAP was not worthwhile, we believe that with joint training we can achieve much better results and is left as a future work.



(a) DCP + CLAHE (32/48 cars detected)

(b) DCPDN (33/48 cars detected)

Figure 6: Sample image comparing object detection performance with DCP + CLAHE vs DCPDN dehazing

4. Experimental Results

In this section, we compare different dehazing + object detection models, provide useful insights about the data-set, discuss different dehazing models on their advantages and disadvantages. Along with these, we have also discussed on evaluation methods and inferences about the dehazing results.

4.1. Datasets

AOD-Net model is trained on the RESIDE training set (ITS+OTS) which is a set of synthetic hazy images. Once we have trained the model, the performance is tested on Real-world Task-driven Testing Set (RTTS) data-set which is a part of RESIDE test-set.

For the the joint AOD-Net + RetinaNet model, training was done using PASCAL VOC dataset with synthetic haze introduced by setting A between $[0.6, 0.8]$ and b between $[0.4, 1.0]$ chosen uniformly in the Eq. 1. The depth parameter estimation was done using methods implemented in the paper[10]

4.2. Training Details

We used Keras to implement the baseline model of AOD-net with 5 convolution layers, 3 concatenation layers. We kept loss function as Mean Squared Error, Learning Rate as 0.001, Decay as 0.0001 and Momentum was set to 0.9 as described in [7]. For one experiment, we loaded pre-trained weights and trained for few epochs; for second experiment, we changed the initializers(TruncatedNormal) for first few convolution layers and retrained only those layers completely. Training was done on OTS and ITS[8] using NVIDIA 1060 GTX GPU. The average PSNR and SSIM values of these experiments are illustrated in Figure 10 and 11.

For the object detection, we borrowed the base code and pre-trained model of Retina-Net from keras-retinanet¹. As the COCO data-set has 80 categories of objects and VOC has only 20, we modified the base code to fit for the VOC data-set requirements as RTTS is organized in the same form as VOC2007. We made changes in the evaluate module as RTTS contains only 5 categories out of those 20 in order to compute the mAP accurately.

Joint model of AOD-Net and Retina-Net was created by modifying the Retina-Net base code to accommodate AOD-Net model in-line with the Retina-Net. This model was trained by unlocking first two layers in AOD-Net on the hazy dataset created from PASCAL VOC as described in the dataset section.

For CLAHE we used OpenCV's CLAHE module and interfaced with our python code. The Dark Channel Prior Dehazing code and pretrained model is borrowed from DCP². The code was extremely inefficient, generating 1 dehazed image in approximately 5 minutes, we were able to get it down to 1 minute by doing certain optimizations. All of our code can be found here³

For the DCPDN[18] implementation, we borrowed the code and pretrained model from DCPDN⁴. The model runs in PyTorch environment we had to change bits and pieces in the code-base as the original code caps the images to 512x512 resolution.

Mask-RCNN base code was borrowed from Mask_RCNN⁵ and pre-built model on COCO dataset⁶ was also taken from the same project. This model was directly used along with different dehazing and image processing techniques (CLAHE +DCP) for the object

detection.

4.3. Comparison with state-of-the-art Methods

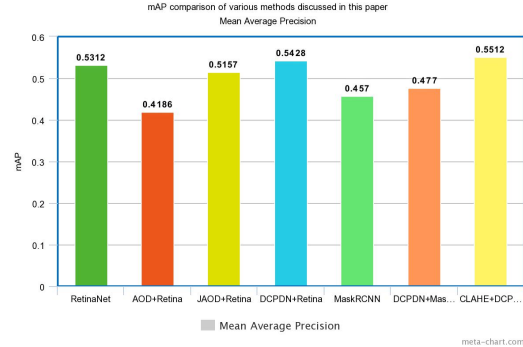


Figure 7: mAP metrics for all the models

As discussed in the previous sections, we have used a bunch of dehazing models to get dehazed images for the object detection. Surprisingly, the baseline model, i.e hazy images + Retina-Net gave us a mAP score of 0.5312. Our expectation was dehazing should provide better results, but as we can see from the Figure 7, few dehazed results with Retina-Net performed poorly in terms of mAP. We attribute this degradation to two factors. First, we retrained the models for less number of epochs. Second, although the dehazing algorithm was increasing the image quality for some of the images, but for others the images were getting darker and objects were masked due to this contrast effect. We believe that this resulted in poor object detection. For example, lets compare the images shown in Figure. 8. Retina-Net model was able to detect motorbike in hazy image but not in AOD-Net dehazed image. We conclude that dehazing might not always be the best way to get better object detection.

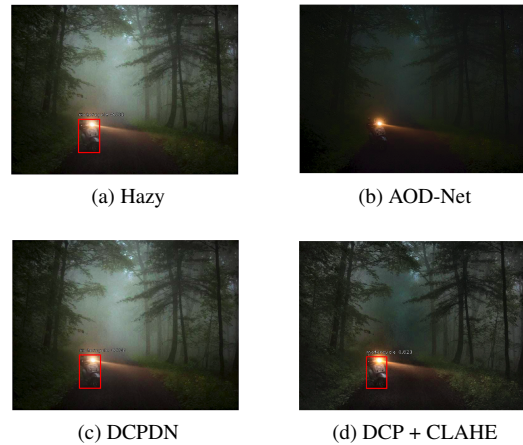


Figure 8: Sample image comparing object detection performance on Hazy, AOD-Net, DCPDN and DCP + CLAHE

¹This is git-hub project maintained by fizyr and available at:<https://github.com/fizyr/keras-retinanet>

²This is a git-hub project by sumitsahrawat and available at : <https://github.com/sumitsahrawat/dcp>

³Project code Google drive link : Submission Code

⁴This is a git-hub project by hezhangsprinter and available at : <https://github.com/hezhangsprinter/DCPDN>

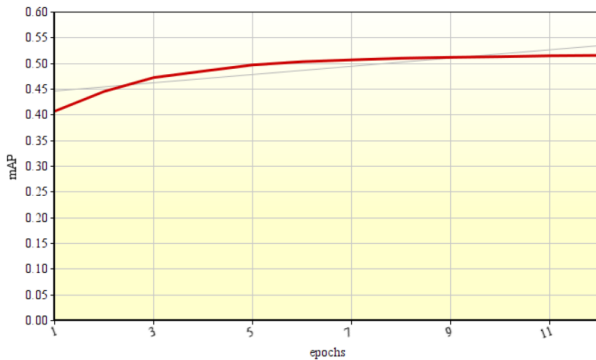
⁵This is a git-hub project by matterport and available at : https://github.com/matterport/Mask_RCNN

⁶<http://cocodataset.org/>

Our new hypothesis was, if AOD-Net model is re-trained along with Retina-net, the model might be able to learn better and accurately represent/re-construct the objects in the images. So, we created a joint model of the AOD-Net and Retina-Net (details mentioned in the previous sections). As expected, this model was able to perform better compared to the earlier disjoint model. We created different model-checkpoints for different epochs. It can be observed from the Figure 9 that mAP was increasing as the number of epochs increased and nearly saturated around 12 epochs (see Figure 9).

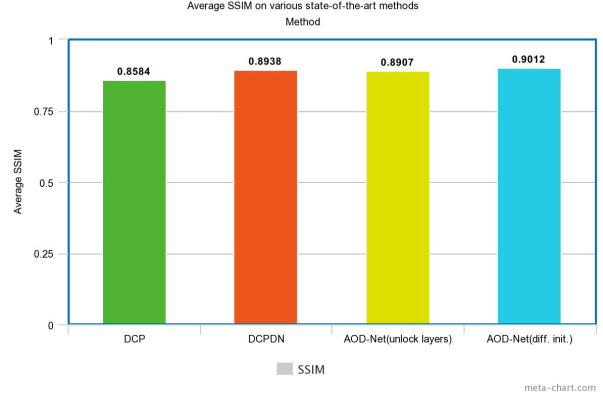
One more inference we can make from the results is that AOD-Net was not able to preserve the edges very well; we could see some dehazed images were losing edges and hence were merging with the background. The DCPDN model was able to preserve the edges in the dehazed images which is essential for the object detection. As expected, with the images processed using DCPDN, we were able to get better mAP of 0.5428 with Retina-Net(see Figure 7).

In terms of number of objects detected and bounding box generation the Mask-RCNN performed very well for the samples of the RTTS data (see Figure 12 and Figure 13). We had our hopes high on getting supreme mAP scores compared to the Retina-Net model. However, we faced some issues with the mAP computation. The details concerned to mAP evaluation of Mask R-CNN are discussed in the next section.



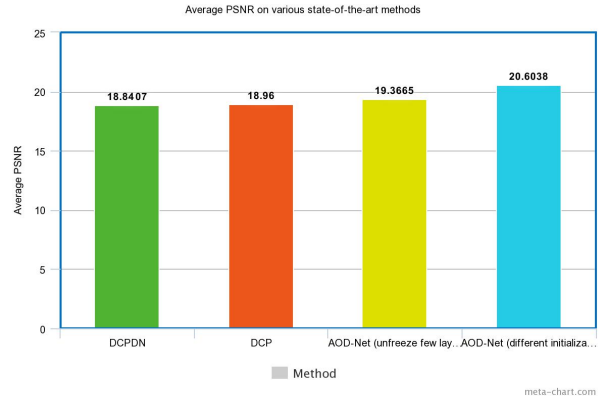
(a) mAP vs epochs

Figure 9: mAP is increasing with epochs in JAOD-Net + Retina-Net model



(a) SSIM vs Models

Figure 10: SSIM for AOD-net is high compared to other models



(a) PSNR vs Models

Figure 11: PSNR for AOD-net is high compared to other models

At last, image processing techniques CLAHE tagged with DCP dehazing was able to get best mAP score of 0.5512 from Retina-Net. Our inference was that, we don't need to have good visual quality of the image for the best detection. As can be seen from the Figures 10 and 11, although AOD-Net performed best when it comes to SSIM and PSNR scores, the mAP for the AOD-Net dehazed results are low. Even though the PSNR and SSIM are comparatively low for DCPDN and CLAHE+DCP, mAP scores for these two dehazed image sets are quite high. We can conclude that, along with image quality there are several properties such as edges and contrast which are essential for the object detection, this is evident and can be observed in Figure 8. In all other instances apart from the AOD-Net, Retina-net was able detect the bike.

4.4. Challenges with mAP Evaluation in Mask-RCNN

Even though MASK-RCNN performed really well in terms of detection on the sample of the RTTS data-set, mAP was comparatively low(Figure 7). Our observation is that the pre-trained model we borrowed is for the COCO dataset which has 80 object categories compared to RTTS which has 5. So, while calculating the mAP, the object detected with higher confidence score which does not belong the 5 categories in RTTS will reduce the mAP as the evaluation considers it as an anomaly.



Figure 12: Ground truth



Figure 13: Mask RCNN

As we can observe from the Figure 12 and Figure 13, the number of cars annotated in Figure 12 (Ground truth) are less than the number of cars detected in Figure 13 (Mask R-CNN output). If the score of the un-annotated object(In Ground truth) is more, this will lead to poor mAP scores, even though the object detection performed better than the ground truth. We are currently working on analyzing the evaluation criteria and how well suited is it for data-set with lesser categories and lesser annotations.

5. Conclusion

The paper analyses several dehazing and object detection algorithms. We tried to improve upon the state-of-the-art

AOD-Net + Faster RCNN as described in [7] with a variety of state-of-the-art object detection methods, on Real-World Task Driven Testing images, using Mean Average Precision criteria. We also presented a study on how AOD-Net can boost the object detection and recognition performance on natural hazy images, by joint pipeline optimization. We also compare our retrained AOD-Net with state-of-the-art AOD-Net[7] using objective (PSNR, SSIM) criteria. Extensive experimental results confirm the superiority, robustness, and efficiency of CLAHE + DCP + RetinaNet. Moreover, it can be observed that the image quality need not be visually appealing in order to enhance the object recognition performance. Although our jointly tuned model did not fare well with CLAHE + DCP + RetinaNet we believe that by further training the joint model and hyper-parameter optimization we can achieve better results. Evaluating a model built to predict multiple classes will tend to give lower mAP on the dataset which has lesser number of classes. So, the modification of evaluation function is one of our future work along with working on joint model of DCPDN and Mask R-CNN.

6. Appendix A: Individual Contributions

Sachin Shreenivas Puranik: Implemented AOD-Net in Keras, Tried various experiments with AOD-Net such as varying filters/Activation functions to enhance the performance(Get the SSIM/PSNR on SOTS), Implemented haze adder module to add haze to the COCO/VOC dataset, Implemented CLAHE+DCP module. Did the initial set-up of Mask-RCNN for the object segmentation. Setup the environment for DCPDN and performed dehazing to get PSNR and SSIM numbers on SOTS. Provided the dehazed results of DCPDN for object detection.

Ramakrishna Prabhu: Understanding Retina-net and necessary modification for VOC PASCAL data-set. Developed Joint AOD-Net and Retina-Net model. Working on evaluation module of MASK RCNN and its joint model of DCPDN. Researched on different models and techniques for object detection. Conversion of Pascal format to COCO format. Handled training and testing of the object detection of Retina, Joint and Mask-RCNN.

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